Assignment 8: SQL Analytical Queries & Views

Overview:

In this assignment, you'll extend your SQL expertise by performing analytical queries on a Products table and creating a view. You will practice aggregate functions, subqueries, ordering, and view creation.

1. Create and Populate Table

```sql
CREATE TABLE Products (
  ProductID INT PRIMARY KEY,
  ProductName VARCHAR(50),
  Price DECIMAL(10,2),
  QuantitySold INT,
  Category VARCHAR(50)
);

INSERT INTO Products (ProductID, ProductName, Price, QuantitySold, Category) VALUES
  (1, 'Laptop',  800.00,  5,  'Electronics'),
  (2, 'Smartphone', 600.00, 10,  'Electronics'),
  (3, 'Desk Chair', 120.00, 15,  'Furniture'),
  (4, 'Table',    200.00,  8,  'Furniture'),
  (5, 'Notebook',         5.00, 20,  'Stationery'),
  (6, 'Pen',      2.00, 50,  'Stationery');
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_1)** and save it in the same **PDF/Doc/Word** file.

2. SQL Query Tasks

1. **a) Above-Average Price:** Select products whose Price exceeds the average price.

SQL:

```
SELECT * FROM Products
WHERE Price > (
  SELECT AVG(Price) FROM Products
);
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2a)** and save it in the same **PDF/Doc/Word** file.

2. **b) Most Expensive Product:** Display the ProductName and Price of the highest-priced item.

SQL:

```
SELECT TOP 1 ProductName, Price

FROM Products

ORDER BY Price DESC;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2b)** and save it in the same **PDF/Doc/Word** file.

3. **c) Category Counts:** Show the total number of products in each Category (using a subquery).

SQL:

```
SELECT DISTINCT Category,
  (SELECT COUNT(*)
   FROM Products P2
   WHERE P2.Category = P1.Category) AS ProductCount
FROM Products P1;
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2c)** and save it in the same **PDF/Doc/Word** file.

4. **d) Below-Average Sales:** Find products with QuantitySold below the average quantity sold.

SQL:

```
SELECT * FROM Products
WHERE QuantitySold < (
  SELECT AVG(QuantitySold) FROM Products
);
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2d)** and save it in the same **PDF/Doc/Word** file.

5. **e) Electronics View:** Create a view for all Electronics products.

SQL:

```
CREATE VIEW ElectronicsView AS
SELECT * FROM Products
WHERE Category = 'Electronics';
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2e)** and save it in the same **PDF/Doc/Word** file.

> Tip: If you receive a "table already exists" error, rename your table (e.g., Products1) before running the CREATE TABLE statement.

Submission Guidelines:

Create One Main Folder (named Assignment8_YourName) which includes -

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment8_YourName.sql**

Convert the folder into ZIP format.
Upload the ZIP folder on the submission portal.

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact the Tutedude mentor.